Exercise 3: The Phillips Curve

Problem Set 4 — Macroeconomics (20285)

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Section 3.1 — The Classic Phillips Curve

Question 1: Backward-Looking Inflation Expectations

Answer: The backward-looking measure $\tilde{E}_t[\pi_{t+1}] = \frac{1}{4}(\pi_{t-1} + \pi_{t-2} + \pi_{t-3} + \pi_{t-4})$ embodies an *adaptive expectations* hypothesis: agents form their forecast of next-period inflation by simply averaging the four most recent quarterly inflation outcomes. Because it is a four-quarter moving average, this series is mechanically smoother than actual inflation — it dampens transitory spikes and responds only gradually to persistent shifts. The measure implicitly assumes that agents use *only* past own-inflation to forecast, ignoring all other potentially informative variables (monetary policy announcements, survey data, commodity prices, etc.). This is a strong assumption, and it means the measure will systematically lag behind turning points in actual inflation and may fail to capture the forward-looking component of expectations that drives pricing decisions.

Question 2: Full-Sample Phillips Curve Scatterplot

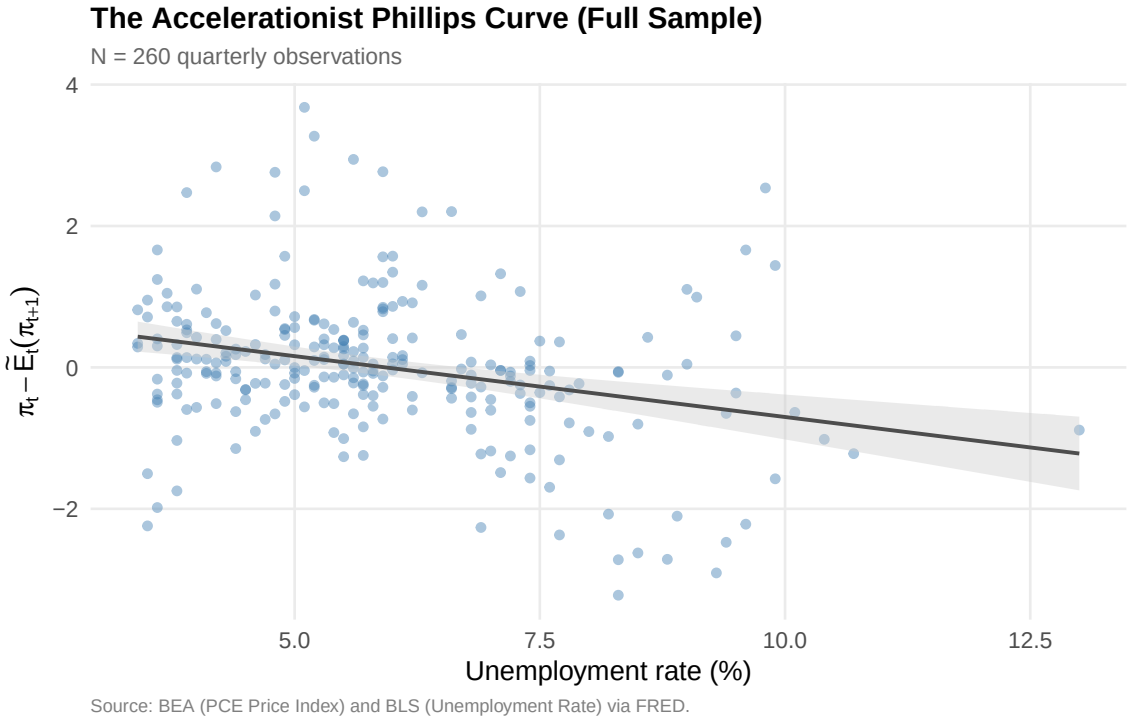

Figure 1: Full-sample accelerationist Phillips curve: inflation surprise vs. unemployment rate.

Answer: The dependent variable $y_t = \pi_t - \tilde{E}_t[\pi_{t+1}]$ measures the *inflation surprise* — the deviation of realised inflation from backward-looking expected inflation. A negative slope coefficient on u_t is consistent with the Phillips curve: when unemployment is high, inflation tends to come in *below* expectations. The full-sample OLS fit ($\hat{\kappa} = -0.172$ (HC1 SE $= 0.044$), $R^2 = 0.081$, $N = 260$) confirms a small but statistically significant negative slope. The economic magnitude is modest and the R^2 low, reflecting the well-known instability of the reduced-form Phillips curve: much of the variation in inflation surprises is driven by supply shocks, expectation shifts, and regime changes that are not captured by a simple bivariate regression on unemployment.

Question 3: Subperiod Phillips Curves

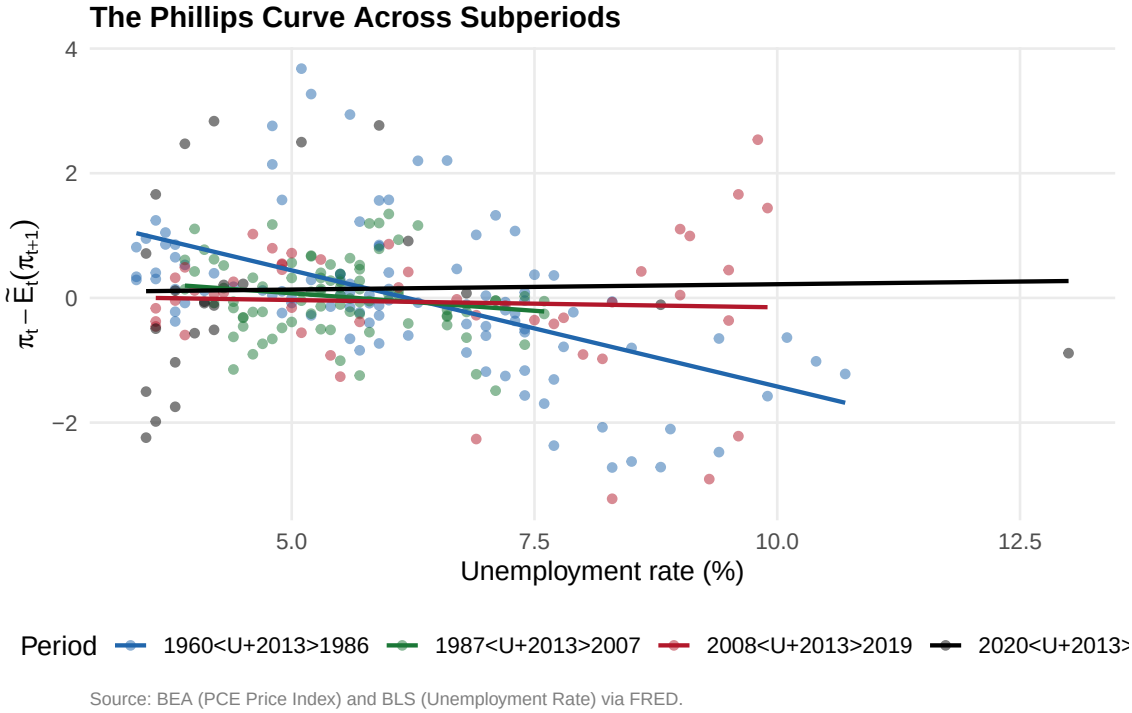

Figure 2: Phillips curve by subperiod, each with its own OLS fit.

Table 1: Phillips-curve slope by subperiod (HC1-robust SEs)

Period	Slope ($\hat{\kappa}$)	HC1 SE	R^2	N
1960<U+2013>1986	-0.373	0.050	0.294	104
1987<U+2013>2007	-0.112	0.067	0.031	84
2008<U+2013>2019	-0.024	0.090	0.002	48
2020<U+2013>2025	0.017	0.116	0.001	24

Answer: The subperiod analysis reveals a striking *flattening* of the Phillips curve over time. During 1960–1986 — a period encompassing both the Great Inflation and the Volcker disinflation — the slope is steeply negative ($\hat{\kappa} \approx -0.37$), reflecting large swings in both unemployment and inflation surprises. The Great Moderation era (1987–2007) shows a much flatter relationship ($\hat{\kappa} \approx -0.11$): well-anchored expectations meant that unemployment fluctuations had diminished pass-through to inflation. The post-GFC period (2008–2019) is flatter still ($\hat{\kappa} \approx -0.02$, $R^2 \approx 0.00$), consistent with the "missing disinflation" puzzle — unemployment rose dramatically during and after the Great Recession, yet inflation barely fell below expectations, suggesting that expectations anchoring (possibly to the Fed's 2% target) rendered the traditional trade-off nearly moot. The COVID era (2020–2025) is harder to read: although the *scatter* spans an extraordinary range — from the 2020:Q2 unemployment spike paired with deflationary surprises to the 2021–2022 inflation surge at very low u_t — the OLS slope is statistically indistinguishable from zero ($\hat{\kappa} \approx 0.02$, HC1 SE ≈ 0.12, $N = 24$),

because supply-driven outliers dominate a short and noisy sample. This pattern is consistent with the idea that the Phillips curve is non-linear and appears flat when expectations are well-anchored and shocks are moderate, but the linear-in-u specification is poorly suited to the COVID episode, where most of the action lies on the steep arm of the slanted-L examined in Question~9.

Section 3.2 — Household Expectations (Coibion & Gorodnichenko 2015)

Question 4: Summary of Coibion & Gorodnichenko (2015)

Answer: The central question addressed by Coibion & Gorodnichenko (2015) is the "missing disinflation" puzzle: why did US inflation not fall much further (or enter deflation) during the Great Recession of 2008–2011, despite the massive spike in unemployment and economic slack? Traditional Phillips curve models, which relied on backward-looking expectations or forecasts by professionals, predicted a much sharper disinflation.

Their key finding is that the Phillips curve is actually "alive and well" once you use the *correct* measure of inflation expectations. They document that household inflation expectations (as measured by the Michigan Survey of Consumers) closely track salient prices, particularly oil and gasoline. During the 2009–2011 period, a surge in global oil prices pushed household inflation expectations higher, even as the broader economy remained in a severe slump. When these survey-based expectations are used in an expectations-augmented Phillips curve, the implied negative inflation surprise ($\pi_t - E_t^{MSC}$) matches the high unemployment rate perfectly. This fully explains the missing disinflation without needing to assume that the underlying slope of the Phillips curve had flattened.

Question 5: Comparing the Two Expectations Series

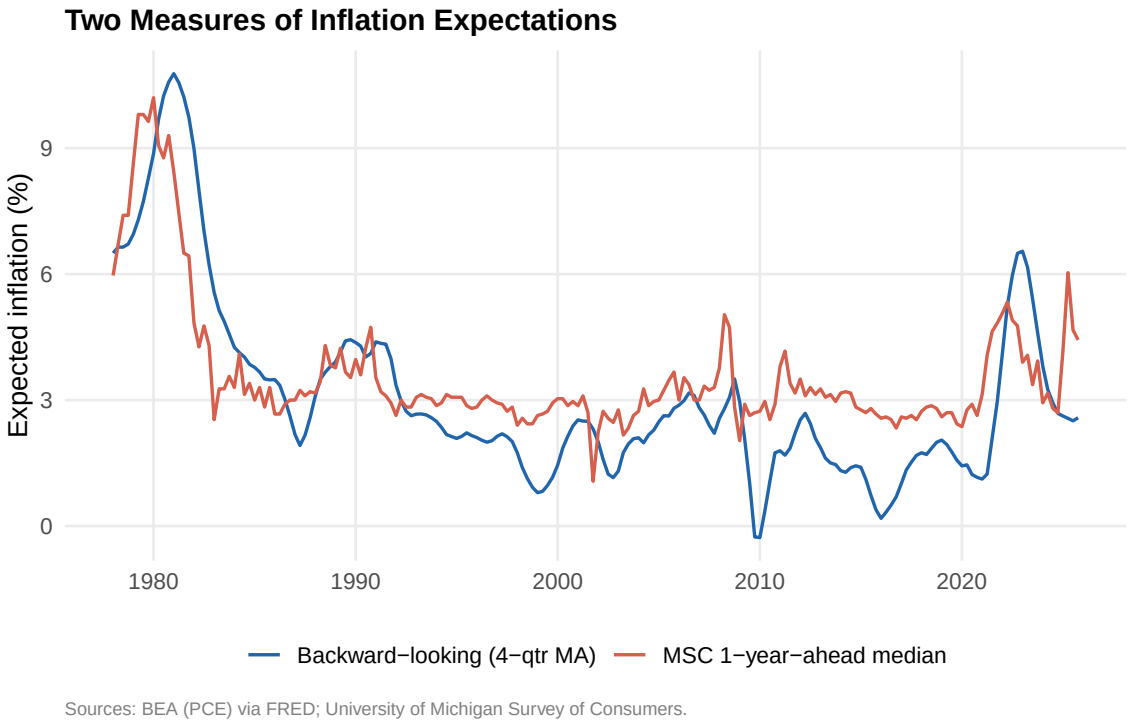

Figure 3: Backward-looking vs. Michigan Survey household inflation expectations.

Answer: Figure @ref(fig:q4-comparison) overlays the two series on a common axis from 1978:Q1 onward. Several key differences emerge between the backward-looking measure $\tilde{E}_t[\pi_{t+1}]$ and the Michigan Survey of Consumers (MSC) 1-year-ahead median expectation $E_t^{MSC}[\pi_{t+4}]$:

- **Level:** The MSC median expectation tends to run *above* the backward-looking measure, particularly during the low-inflation period after 2000. Households systematically perceive and expect higher inflation than is recorded by the PCE index, a well-documented finding in the survey literature.
- **Volatility:** The backward-looking measure, being a four-quarter moving average of actual inflation, is mechanically smooth. The MSC median is somewhat more volatile and can move abruptly in response to salient price signals (e.g., gasoline prices).
- **Co-movement with actual inflation:** Both series co-move with actual inflation at low frequencies, but the MSC expectations diverge notably during the post-GFC period — they *rose* in 2009–2011 even as actual (PCE) inflation was falling, a key insight exploited by Coibion & Gorodnichenko (2015).

The *median* (rather than the mean) is preferred for household expectations because survey response distributions are typically right-skewed and fat-tailed. A small fraction of respondents consistently report extreme values (e.g., 20–50% expected inflation), which disproportionately inflate the mean. The median is robust to these outliers and provides a more representative summary of the central tendency of household beliefs.

Question 6: Phillips Curve with MSC Expectations

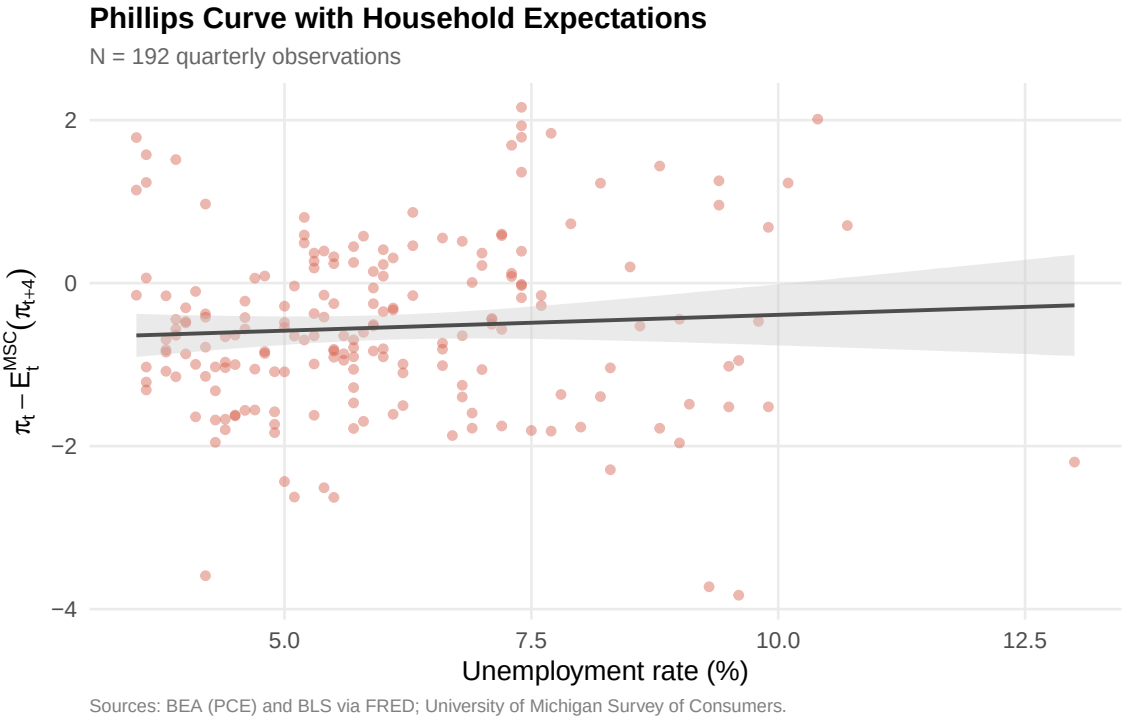

Figure 4: Phillips curve using MSC household expectations (contemporaneous specification).

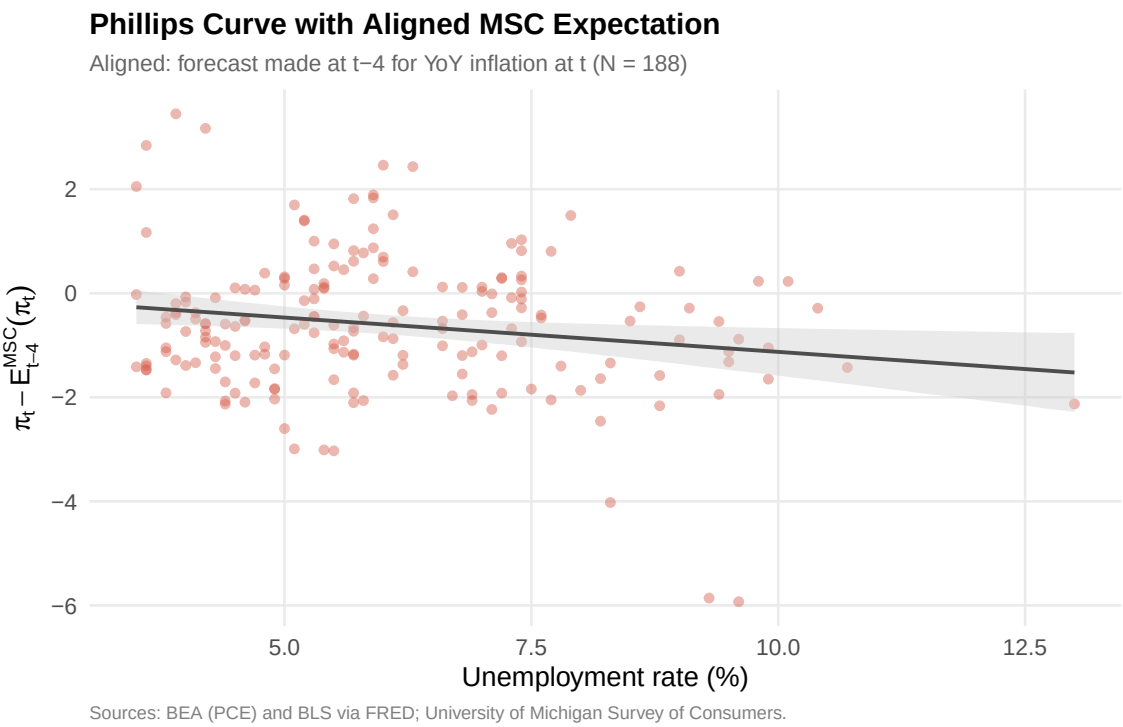

Figure 5: Phillips curve with the MSC forecast aligned to the realisation it predicts: $\pi_t - E_{t-4}^{MSC}[\pi_t]$.

Answer: When household expectations from the Michigan Survey of Consumers are used instead of backward-looking expectations, the Phillips curve relationship changes markedly compared to Question 2. With the *contemporaneous* specification $y_t = \pi_t - E_t^{MSC}[\pi_{t+4}]$ the OLS slope is small, of indeterminate sign, and statistically indistinguishable from zero ($\hat{\kappa} = 0.039$ (HC1 SE $= 0.058$), $R^2 = 0.004$, $N = 192$); the scatterplot is essentially flat.

This result has both an economic and a measurement component. First, household inflation expectations — particularly since the Global Financial Crisis — tend to be *biased upward* relative to realised inflation, so $\pi_t - E_t^{MSC}[\pi_{t+4}]$ is negative on average over the sample. Second, the dependent variable as constructed is *not a properly aligned conditional surprise*: π_t is current year-on-year inflation, while $E_t^{MSC}[\pi_{t+4}]$ is a forecast for inflation four quarters ahead. The realisation and the forecast horizon do not match.

A simple diagnostic confirms that the timing convention matters. Substituting $E_{t-4}^{MSC}[\pi_t]$ — the forecast made one year earlier of the YoY inflation rate that has just realised — recovers a downward-sloping relationship ($\hat{\kappa} = -0.132$ (HC1 SE $= 0.059$), $R^2 = 0.032$, $N = 188$). The slope flips from a small positive to a small but visibly negative number once the forecast is dated correctly, so part of the "missing-disinflation flattening" in the contemporaneous specification is mechanical mis-alignment rather than evidence on the structural Phillips slope.

These findings are consistent with the central insight of Coibion and Gorodnichenko (2015). When household expectations are used, the phenomenon of "missing disinflation" essentially

disappears. During the 2009–2013 period, household inflation expectations rose sharply (driven by oil and gasoline prices), even as the economy was in a deep recession. Because E_t^{MSC} was high, the inflation surprise was already very negative — exactly what the Phillips curve would predict when unemployment is elevated. The "puzzle" arose only when backward-looking or professional-forecaster expectations (which fell during the recession) were used, generating a small negative surprise inconsistent with the large unemployment gap.

Question 7: Simultaneity Bias and OLS Identification

Answer: A fundamental identification problem afflicts all the OLS regressions estimated above (in Questions 2, 3, and 6): **simultaneity (reverse causality)**. Inflation expectations and unemployment are *jointly determined* — aggregate demand shocks simultaneously raise inflation (or expectations thereof) and lower unemployment, while monetary policy responses to inflation and employment conditions shift the economy along both dimensions. OLS of inflation on unemployment therefore conflates movements *along* the Phillips curve (driven by supply shocks, which we want to identify) with shifts *of* the Phillips curve (driven by demand shocks and policy responses).

McLeay & Tenreyro (2020, *NBER Macroeconomics Annual* 34) formalise this argument. They show that when the central bank conducts *optimal monetary policy* — adjusting interest rates to stabilise inflation in response to unemployment gaps — the policy reaction function induces a spurious positive correlation between inflation and unemployment in equilibrium. This endogeneity biases the OLS estimate of the Phillips curve slope toward zero (or even toward a positive sign). The PC is identified *only* off supply-side shocks or when an instrumental variable for monetary policy is available.

Relating this back to our estimates:

- **Question 2 (Full sample, backward-looking):** The near-zero slope is *most* susceptible to this bias. Backward-looking expectations move slowly and essentially proxy for lagged inflation, so the regression captures a mix of demand- and supply-driven variation, with the Fed's stabilisation policy washing out much of the PC signal.
- **Question 3 (Subperiods):** The subperiod estimates are similarly biased. The observed "flattening" during the Great Moderation (1987–2007) may partly reflect a more aggressive inflation-targeting regime by the Fed, which offset demand shocks more effectively, rather than a true structural flattening of the underlying Phillips curve.
- **Question 6 (MSC expectations):** While somewhat less biased because household survey expectations capture *forward-looking* information that partially controls for demand-shock-induced expectation shifts, simultaneity remains. Demand booms raise both MSC expectations and employment, confounding the true slope estimate.
- The only clean identification requires either an *instrument* for monetary policy (e.g., narrative shocks à la Romer & Romer) or an instrument for slack (e.g., Bartik-style local labour-market instruments, as in Gitti 2024).

Section 3.3 — Non-linear Phillips Curves

Question 8: Summary of Benigno & Eggertsson (2024)

Answer: Benigno & Eggertsson (2024) argue that the standard *linear* New Keynesian Phillips curve misses a critical feature of the inflation–labour-market-tightness relationship: it is **non-linear, exhibiting a convex (or kinked) shape**. Specifically, they document a "slanted-L" pattern in the data:

- **At low tightness** (high unemployment, low V/U), additional slack has *little* disinflationary effect — the Phillips curve is nearly flat. This explains the "missing disinflation" of 2009–2015: despite very high unemployment, inflation did not fall substantially because the economy was on the flat portion of the curve.
- **At high tightness** (low unemployment, high V/U), small increases in tightness generate *large* inflation increases — the curve becomes steep, nearly vertical. This explains the rapid inflation surge of 2021–2022: the economy quickly moved onto the steep portion of the curve.

The key theoretical mechanism is a *capacity constraint on production*: firms can substitute between vacancies and labour to adjust output, but as the labour market approaches full utilisation the marginal cost curve becomes very steep — effectively kinking — because additional output requires either bidding scarce workers away from other firms or accepting longer vacancy durations. Once this constraint binds, small increases in tightness map into large increases in marginal cost and hence in price-setting markups, generating the near-vertical arm of the slanted-L. Below the kink, by contrast, marginal cost is roughly flat in tightness, so additional slack has little disinflationary bite. Search-and-matching frictions are a complementary force: when V/U is high, hiring takes longer and wage offers must rise to attract workers, reinforcing the cost-side amplification. Crucially, Benigno & Eggertsson argue that *no structural break or shift in the Phillips curve is needed* to explain both the "missing disinflation" and the "inflation surge" — both phenomena arise naturally from the non-linear shape of a *stable* underlying relationship.

Question 9: Replicating Benigno & Eggertsson (2024) Figure 1

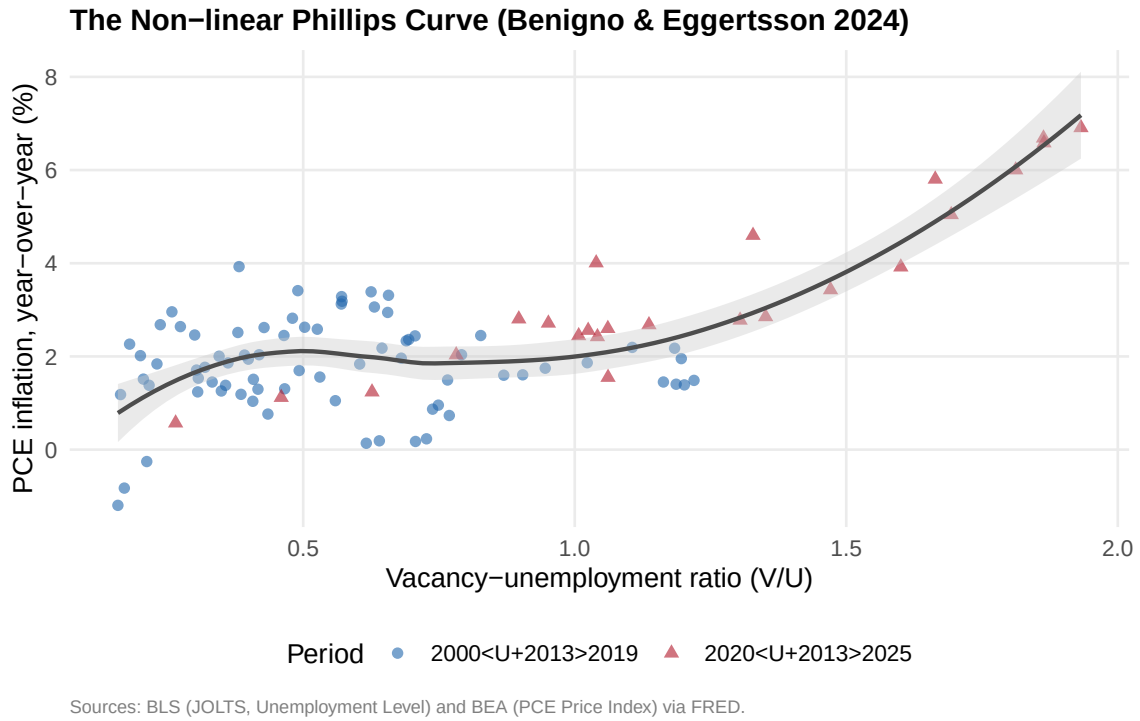

Figure 6: Non-linear Phillips curve: PCE inflation vs. labour-market tightness (V/U).

Answer: The scatterplot reveals the characteristic **"slanted-L" shape** described by Benigno & Eggertsson (2024). For V/U ratios below roughly 1, the relationship is essentially flat: across the 2000–2019 expansion, the GFC trough, and the late-cycle 2018–2019 quarters where V/U first crossed above one with inflation still anchored around 2%, year-over-year PCE inflation hovers in a narrow 1–2.5% band regardless of tightness. The steep, near-vertical arm of the curve is essentially the **2021:Q2–2023:Q1 cluster**: every quarter in our sample with PCE inflation above 4% sits in this window, with V/U ratios between 1.04 and 1.93. Within this cluster a roughly 0.9-point increase in V/U (from 1.04 to 1.93 over 2021:Q2–2022:Q2) is associated with PCE inflation rising from 4.0% to 6.9%. The kink itself sits near $V/U \approx 1$ — close to where Benigno & Eggertsson place it in their Figure 1.

This pattern contrasts sharply with the **linear Phillips curve**, which would predict a uniform, moderate negative relationship between slack and inflation across the entire range. The linear specification would substantially overpredict disinflation during the slack post-GFC period and underpredict inflation during the 2021–2022 boom. The non-linear relationship shows that inflation is relatively insensitive to tightness when the labour market has ample slack — firms can fill vacancies without large wage increases — but becomes highly sensitive once tightness pushes the economy past the kink and onto the capacity-constrained arm. As the post-2022 observations show, the unwinding of tightness brought inflation back down, tracing the curve in reverse: the 2024–2025 quarters are back at $V/U \approx 1$ with PCE inflation around 2.5%.

Question 10: Summary of Gitti (2024)

Answer: Gitti (2024) addresses a key **endogeneity concern** that plagues the OLS scatterplot in Question 9: the vacancy-unemployment ratio V/U is endogenous to inflation because demand shocks simultaneously raise both labour-market tightness and prices. Her identification strategy uses a **Bartik-style (shift-share) instrumental variable**: she exploits the fact that local labour markets are differentially exposed to national industry-level vacancy shocks. By interacting pre-determined local industry employment shares (the "shares") with national industry-level vacancy changes (the "shifts"), she constructs an instrument that captures exogenous variation in local tightness that is plausibly uncorrelated with local demand shocks.

This approach relates directly to Question 9 (and Question 7): the OLS scatterplot is subject to the **same simultaneity bias** discussed previously. During demand booms, both tightness and inflation rise *simultaneously*, inflating the apparent slope of the tightness–inflation relationship. Conversely, during recessions, both decline together. The OLS estimates therefore confound the causal effect of tightness on inflation with the common response of both variables to demand conditions.

Gitti's IV strategy recovers a **cleaner causal estimate** of the tightness–inflation relationship, which she finds to be:

1. **Non-linear**, consistent with the slanted-L shape documented by Benigno & Eggertsson (2024);
2. **Quantitatively different** from OLS — the IV estimates suggest the true causal relationship is steeper in the tight-market region than OLS implies (because OLS averages over demand- and supply-driven variation);
3. **Robust** to controlling for local demand conditions and to various specification checks on the instrument's validity.

Her results thus provide rigorous causal support for the non-linear Phillips curve hypothesis and demonstrate that naive OLS estimates — whether using unemployment (Q2, Q5) or tightness (Q8) — are biased by simultaneity.

Appendix

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Running under: macOS 26.3.1

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